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RESEARCH ARTICLE

Investigating the subjective and objective efficacy of a cognitive behavioural therapy for insomnia (CBT-I)-based smartphone app on sleep: A randomised controlled trial

Alexandra Hinterberger[1] | Esther-Sevil Eigl[1] | Robyn Nina Schwemlein[1] | Pavlos Topalidis[1] | Manuel Schabus[1,2]

[1]Laboratory for Sleep, Cognition & Consciousness Research, University of Salzburg, Salzburg, Austria

[2]Centre for Cognitive Neuroscience Salzburg (CCNS), University of Salzburg, Salzburg, Austria

Correspondence
Manuel Schabus, Laboratory for Sleep, Cognition & Consciousness Research, University of Salzburg, A-5020 Salzburg, Austria.
Email: manuel.schabus@plus.ac.at

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Summary

Due to insufficient treatment options for insomnia, effective solutions are urgently needed. We evaluated the effects of a CBT-I-based app combining sleep training with subjective and objective sleep monitoring on (i) sleep and (ii) subjective-objective sleep discrepancies (SOSD). Fifty-seven volunteers (20–76 years; $M_{Age} = 45.67 \pm 16.38$; 39 female) suffering from sleep problems were randomly assigned to an experimental group (EG, $n = 28$) or a waitlist control group (CG, $n = 29$). During the 6-week app phase, the EG used the CBT-I-based programme and a heart rate sensor for daily sleep monitoring and -feedback, while the CG used sleep monitoring only. Sleep was measured (i) subjectively via questionnaires (Insomnia Severity Index, ISI; Pittsburgh Sleep Quality Index, PSQI), (ii) objectively via ambulatory polysomnography (PSG), and (iii) continuously via heart-rate sensor and sleep diaries. Data revealed interactions for ISI ($p = 0.003$, $\eta^2_{part} = 0.11$) and PSQI ($p = 0.050$, $\eta^2_{part} = 0.05$), indicating training-specific improvements in EG, yet not in CG. While PSG-derived outcomes appear to be less training-specific, a tendential reduction in wake after sleep onset (WASO) was found in EG ($p = 0.061$, $d = 0.55$). Regarding changes in SOSD, the results indicate improvements during the app phase (EG) for sleep efficiency, sleep onset latency, and WASO ($p \leq 0.022$, $d \geq 0.46$); for total sleep time both groups showed a SOSD reduction. The findings indicate beneficial effects of a novel smartphone app on sleep and SOSD. More scientific evaluation of such digital programmes is needed to ultimately help in reducing the gap in non-pharmacological insomnia treatment.

KEYWORDS
app, CBT-I, dCBT, digital programme, insomnia, mHealth

1 | INTRODUCTION

According to Ohayon (2002) approximately one third of the general population is experiencing dissatisfaction with the quantity and/or

Alexandra Hinterberger and Esther-Sevil Eigl contributed equally to this study and share first authorship.

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quality of sleep, while insomnia, which is characterised by difficulty initiating and/or maintaining sleep and/or early awakenings and resulting daytime impairments as defined by the ICD-11 (World Health Organization, 2019) and the Diagnostic and Statistical Manual of Mental Disorders (DMS-5; American Psychiatric Association, 2013), is considered to be the most common sleep disorder in western society, affecting about 6–15% of the adult population (Ohayon, 2011).

Clinical guidelines (e.g., Riemann et al., 2017) recommend cognitive behavioural therapy for insomnia (CBT-I) as the first line of treatment. CBT-I is a non-pharmacological, well-established measure that improves sleep outcomes persistently, using components such as relaxation techniques, sleep restriction, stimulus control, psychoeducation, and cognitive therapy (e.g., Edinger & Means, 2005; Mitchell et al., 2012; Okajima et al., 2011). Due to limited access, a lack of adequate referral from primary care providers or missing compliance from patients, however, most individuals affected do not receive first-line treatment (Koffel et al., 2018).

Digital CBT-I (dCBT-I), that is, the implementation of CBT-I contents in websites or smartphone applications, has been studied extensively in recent years (e.g., Soh et al., 2020; Vedaa et al., 2019) and it has been shown to generate almost comparable effects to face-to-face (F2F) CBT-I, while also being more cost effective (Luik et al., 2017). Using smartphone applications to convey this type of treatment seems to be the most convenient way (Khan et al., 2015). There is a broad body of research concerning the impact of dCBT-I on subjective measures of sleep, showing durable improvements in sleep quality, insomnia severity, self-reported sleep efficiency (SE), wake after sleep onset (WASO) as well as sleep onset latency (SOL) (Espie et al., 2012; Horsch et al., 2017; Lorenz et al., 2019; Ritterband et al., 2017). A network meta-analysis by Simon et al. (2023) further revealed medium effect sizes regarding the improvement of insomnia severity through both guided and unguided dCBT-I programmes.

Objective measures, however, have rarely been considered in intervention studies. Despite insomnia being defined as a psychological disorder, physiological changes in sleep (e.g., reduced SE, prolonged SOL, increased WASO, less total sleep time [TST]) can be found in the majority of insomnia patients in comparison to healthy controls (Baglioni et al., 2014; Edinger & Krystal, 2003), which makes an investigation of these variables in the context of sleep interventions highly relevant.

Indeed, while PSG-derived changes in sleep are less frequently observed compared to subjective changes (as noted by Mitchell et al., 2019), objective sleep measurement can still play a crucial role in tracking relevant changes and improvements during CBT-I. For instance, Simon et al. (2023) reported small, yet positive, effects for F2F CBT-I regarding objective SOL, as well as moderate negative effects on TST, which appears to be an important objective measure regarding insomnia (Scott et al., 2022). Analysing studies using actigraphy or PSG data, Okajima et al. (2011) ascertained medium to large effect sizes for changes in the total wake time, early morning awakening, WASO, and SE at post-treatment, as well as for SOL at follow-up. Moreover, Krystal and Edinger (2010) discovered improvements in objectively measured WASO and SE by examining the impact of F2F CBT-I with ambulatory polysomnography (PSG) implementation in contrast to a placebo control group. Similarly, Cervena et al. (2004) reported improvements in several objective sleep parameters (i.e., TST, SE, WASO) with medium to large effect sizes after individually delivered CBT-I. Regarding sleep microstructure, reduced beta and sigma activity was observed in response to the treatment, which suggests that CBT-I might also improve deviations on physiological markers of hyperarousal. To summarise, investigating objective measures of sleep can provide deeper insight in intervention-specific effects, as shown for F2F CBT-I. For dCBT-I, research on objective measures of sleep remains sparse.

A paradoxical phenomenon, frequently reported in insomnia sufferers, which might be based on physiological hyperarousal (e.g., increased micro-arousals, beta- and gamma activity; Perlis et al., 2001; Riemann et al., 2010), is a discrepancy between the subjective perception of sleep and objectively derived measures (subjective–objective sleep discrepancy; SOSD). The fact that an underlying hyperarousal causes the brain to remain in an overactive state, may in consequence lead to worse subjective perception of sleep compared to objectively measured sleep. Hence, the typical image of SOSD is the underestimation of TST, while SOL and WASO are overestimated (Carskadon et al., 1976; Frankel et al., 1976). As SOSD can frequently be observed in individuals suffering from insomnia and may not only impair daytime functioning (Semler & Harvey, 2005), but may also potentially lead to patients failing to recognise therapy-related improvements in general or to become more anxious about their sleep (Crönlein et al., 2019; Tang & Harvey, 2004), it is necessary to address this deviant perception of sleep. In line with findings by Cervena et al. (2004), previous studies have reported that F2F CBT-I leads to considerable improvements of SOSD in several parameters (i.e., WASO, TST, SOL; Crönlein et al., 2019; Lund et al., 2013; Nishikawa et al., 2021). However, to our knowledge, no study has investigated whether SOSD improves using app-based dCBT-I.

To keep track of abnormal sleep patterns and possible harmful secondary damage, such as a higher risk of affective disorders (Baglioni et al., 2011), but also developing cardio-vascular diseases (Sofi et al., 2012), type 2 diabetes (Ayas et al., 2003), or obesity (Ogilvie & Patel, 2017), it is necessary to investigate the effects of (d) CBT-I not only on subjective, but also on objective measures of sleep. Consequently, an integration of objective measures in the investigation of the impact of (d)CBT-I may contribute to a better understanding of insomnia-related complaints per se, as well as the effects of treatment on sleep, especially since research in this field is still scarce (Mitchell et al., 2019).

As the conduction of PSG derivations, however, is time-, cost- and resource-intensive, alternative scientifically validated and reliable options for continuous sleep monitoring are urgently needed. In this context, a novel, evidence-based smartphone application (NUKKUAA®), including a fully automated sleep training programme in combination with a state-of-the-art sleep monitoring technology (Topalidis, Baron, et al., 2023; Topalidis, Heib, et al., 2023), achieving near to human-expert classification accuracies, has been developed. While in a first study, the aim was to investigate the impact of this application on subjective sleep improvements and the alleviation of

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comorbid clinical symptoms (i.e., depression and anxiety; Eigl et al., 2023), here, we focus on effects regarding both subjective and objective sleep parameters.

2 | METHODS

2.1 | Participants

Participants have been broadly recruited through public media, adverts in newspapers, and social media. An online pre-screening based on questionnaires and subjective statements was performed prior to enrolment in the study: Inclusion criteria were being 18+ years old, having access to a smartphone, and a minimum score of 8 on the Insomnia Severity Index (ISI; Dieck et al., 2018), indicating at least sub-threshold insomnia, or a minimum score of 6 on the Pittsburgh Sleep Quality Index (PSQI; Riemann & Backhaus, 1996), indicating at least poor sleep quality. Exclusion criteria, assessed through participants' own statements in the pre-screening, were working (night-)shifts, suffering from severe psychological and/or neurological disorders, and/or suffering from a sleep disorder other than insomnia. Intake of (sleep) medication was not defined as an exclusion criterion, however, it was assessed prior to enrolment in the study and thoroughly documented by participants over the course of the study. Initially, 66 people were acquired to take part in this study. Nine participants dropped out during the study, resulting in a dropout rate of 13.6%, which is significantly less than in similar studies investigating the effects of digital programmes for insomnia in outpatients (40% in Peter et al., 2019; 34.4% in Yeung et al., 2015). A total of 57 volunteers (39 female) aged 20–76 years ($M = 45.67 \pm 16.38$), were included in the final sample. The sample was highly educated with 59.6% ($n = 34$) participants stating that they had a university degree. Most participants reported at least moderate sleep problems. At pre-screening participants reported ISI-scores ranging from 6 to 27 ($M = 16.44 \pm 4.51$), with 71.9% of the subjects reporting a score of at least 15, indicating moderately severe clinical insomnia, and PSQI-scores between 4 and 18 ($M = 10.07 \pm 3.34$), with 57.9% of subjects reporting a score of at least 10, indicating chronic sleep problems.

2.2 | Study design and procedure

A randomised controlled trial (RCT) with two parallel arms was conducted: an app-based CBT-I group (experimental group; EG), and a waitlist control group (CG). The flowchart (cf. Figure 1) shows the recruitment process, randomisation, and main assessment time points. Randomisation was guaranteed by using the Research Randomiser tool (Urbaniak & Plous, 2013). At randomisation, that is, the beginning of the app phase, the EG and CG did not differ in age, sex, medication intake, ISI scores, PSQI scores, PSG measures (except for time in bed (TIB)), and discrepancy measures (cf. Table 1).

Main assessments were conducted at four time points: the start of the study (T0), the beginning (T1), and the end (T2) of the 6-week app phase as well as a follow-up assessment 4 weeks after the end of the app phase (T3). At T0–T2, questionnaires were administered online, and ambulatory PSG-data as well as data from a heart rate sensor (HR sensor; Polar® H10, Polar Electro; Kempele, Finland) were collected. Furthermore, at T1 and T2 data from the Polar® Verity Sense, an optical pulse sensor, were gathered. Four weeks later, questionnaire data were collected as a follow-up assessment (T3; cf. Figure 2).

The 2 weeks between T0 and T1 served as a baseline before entering the app phase with participants being randomly assigned to one of the two groups (EG, CG). Procedures were kept consistent for all participants until T1. Afterwards, the EG completed a 6-week sleep training using the full version of NUKKUAA®, which included wearing the HR sensor and receiving feedback about their last night's sleep. The CG, on the other hand, used a reduced version of the app, which only allowed for accessing the sleep diaries and recording of objective sleep data using the HR sensor without providing participants with any feedback on their sleep data or contents. Participants of the CG received full access to the app for 6 weeks upon completion of the follow-up assessment.

The study was approved by the ethical committee of the University of Salzburg (GZ 46/2020) and was conducted in accordance with the Declaration of Helsinki. All participants had given informed consent prior to enrolment in the study protocol.

2.3 | CBT-I-based app programme

NUKKUAA® was developed as a fully automated sleep training application combined with objective sleep monitoring employing heart rate variability measured via HR sensor, and reliably classifying sleep epoch-by-epoch into four stages (Topalidis, Baron, et al., 2023; Topalidis, Heib, et al., 2023). The sleep training is based on a manual to deliver CBT-I (Scharfenstein & Basler, 2004), presented in German language, and consists of a total of six chronologically presented levels, with the requirement of completing the previous level before being able to access content of the consecutive level. Each level consists of the following elements: an information-video providing basic sleep education, an automated chat-bot encouraging cognitive restructuring, audio-relaxation exercises, sleep hygiene advice as well as two blogs covering essential scientific information about sleep. Behavioural components such as sleep restriction and stimulus control are implemented in the chat-bot, advice on sleep hygiene, and the daily sleep recommendations. At the end of each level, a progress reflection regarding the current sleep estimation is unlocked. For a more detailed overview on the app programme, please see Table S1. In order to advance to the next level, every exercise must be completed at least once, while relaxation exercises must be completed at least 7 times (i.e., every day in the present study, as completion is counted 1× a day), and a sleep diary at least 5 times. Besides the exercises, a sleep diary consisting of 5–6 questions is part of the training programme. Participants are reminded once a day through push-notifications to fill out the sleep diary (if not completed) and to continue their training.

After reading out the HR sensor data in the morning, participants are automatically forwarded to fill in their sleep diary and are subsequently

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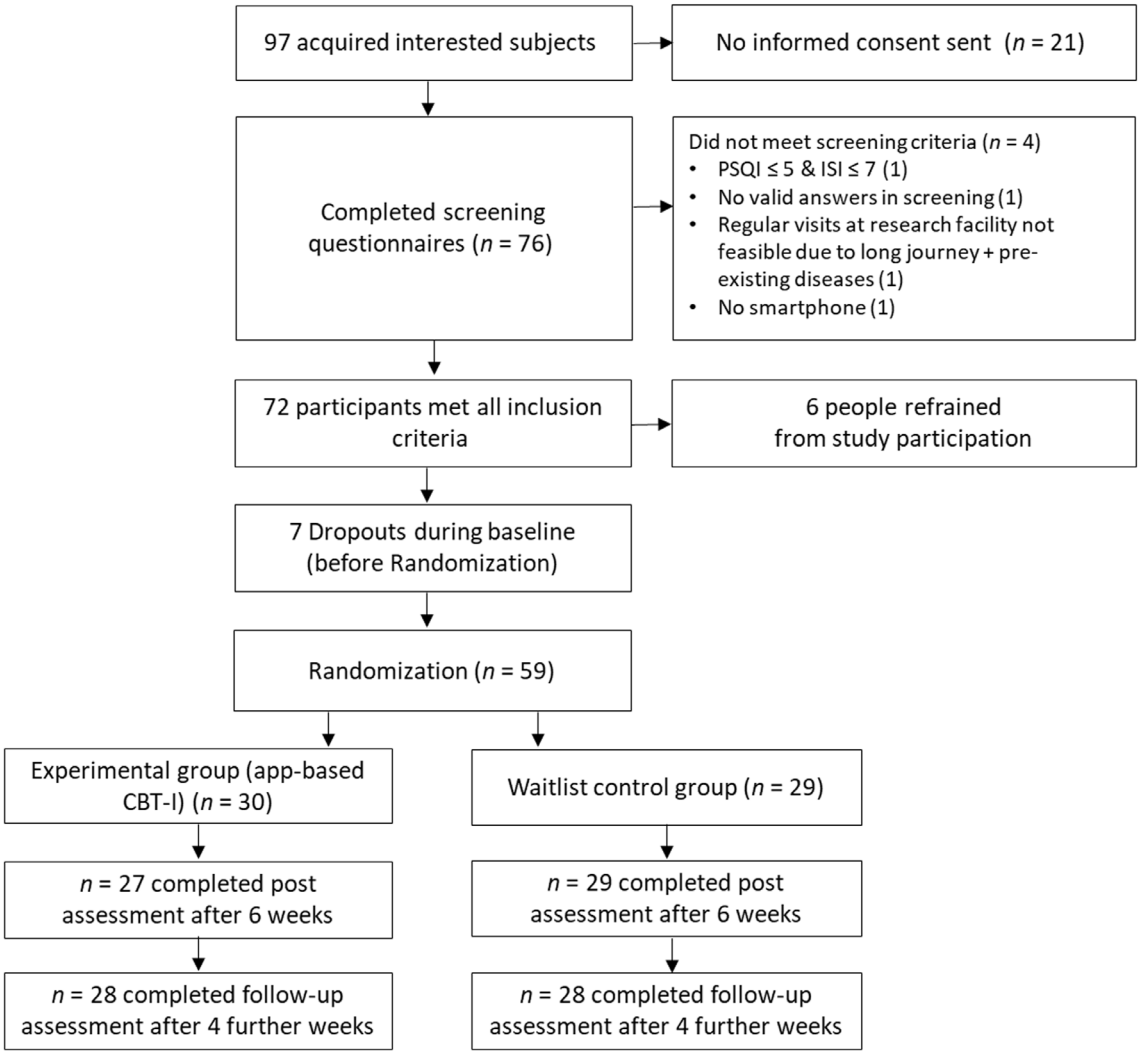


FIGURE 1 Flow of participants.

able to have an overview of the data collected (i.e., subjectively as well as objectively measured sleep parameters and the hypnogram of the last night plus weekly summary). In order to help users understand these results, comparative data from the Siesta Database (Anderer et al., 2005) is provided showing data of healthy individuals of the same sex in the same age category.

During the 6-week app phase investigated in this study, participants of the EG were instructed to complete the programme's first six levels, which are designed to contain the most important elements of CBT-I. In contrast to the EG, the CG did not have access to the sleep feedback after filling out the sleep diary during the app phase.

2.4 | Material and measures

2.4.1 | Subjective sleep assessment

In order to gain subjective measures of sleep, two questionnaires were administered at T0, T1, T2, and T3. They included German versions of the PSQI and the ISI. Furthermore, digital sleep diary data were collected, whenever the participant tracked their sleep using NUKKUAA®.

2.4.2 | Objective sleep assessment

*Objective in-app sleep measurement with a **HR** sensor*

Objective sleep was recorded via NUKKUAA®, using the Polar® H10 (Polar Electro GmbH Deutschland), a HR sensor worn as a chest belt which is comparable in quality to classical two-channel electrocardiograms. Data were processed using the deep learning network as described in Topalidis, Baron, et al. (2023) and Topalidis, Heib, et al. (2023).

2.4.3 | Ambulatory polysomnography

To obtain reliable objective sleep data, a polysomnographic assessment was carried out using the *varioport* ambulatory electroencephalography (EEG) system with 16 channels (Becker Meditec®) in conjunction with gold cup electrodes (Grass Technologies, Astro–Med GmbH®).

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TABLE 1 Sample characteristics at time point of randomisation.

	Experimental group ($n = 28$) M ± SD	Control group ($n = 29$) M ± SD	
Age	48.23 ± 16.99	43.34 ± 15.56	$t(57) = 1.15, p = 0.254$
Sex	8 male, 22 female	12 male, 17 female	$\chi^2(1) = 1.42, p = 0.233$
Medication (overall)	13 yes/15 no	11 yes/18 no	$\chi^2(1) = 0.42, p = 0.516$
Psychoactive medication	3 yes/25 no	6 yes/23 no	$\chi^2(1) = 1.07, p = 0.302$
Sleep medication	2 yes/26 no	1 yes/28 no	$\chi^2(1) = 0.39, p = 0.532$
ISI scores	14.57 ± 5.02	14.75 ± 5.13	$t(56) = -0.14, p = 0.891$
PSQI scores	8.87 ± 3.83	9.64 ± 3.03	$t(56) = -0.85, p = 0.398$
PSG measures			
Total sleep time (min)	417.31 ± 65.98	396.53 ± 57.94	$t(53) = 1.11, p = 0.271$
Time in bed (min)	510.41 ± 74.25	463.57 ± 59.61	$t(53) = 2.20, p = 0.032^*$
Sleep efficiency (%)	83.55 ± 10.52	85.65 ± 6.71	$t(53) = -1.12, p = 0.270$
Sleep onset latency (min)	17.58 ± 13.67	13.58 ± 8.21	$t(53) = 0.41, p = 0.681$
Wake after sleep onset (min)	55.82 ± 40.50	45.94 ± 30.40	$t(53) = 1.46, p = 0.149$
Discrepancy measures			
Total sleep time (min)	13.63 ± 56.93	10.04 ± 32.81	$t(55) = 0.45, p = 0.658$
Sleep efficiency (%)	4.08 ± 12.86	2.21 ± 8.28	$t(55) = 0.80, p = 0.428$
Sleep onset latency (min)	4.18 ± 17.30	4.97 ± 20.90	$t(55) = 0.62, p = 0.537$
Wake after sleep onset (min)	−7.78 ± 40.97	−12.08 ± 19.11	$t(35.44) = 0.92, p = 0.364$

Note: Baseline differences between groups at randomisation. Medication (overall) includes psychoactive medication, sleep medication, and other medication (e.g., for cardiovascular diseases); psychoactive medication includes sleep medication and other psychoactive medications (e.g., antidepressants); sleep medication includes prescribed sleep medication. Discrepancy measures are based on subjective (sleep diary-derived) and objective (heartrate sensor-derived) sleep measures in the first week of the 6-week app phase. Statistically significant comparisons are marked with a star (*).
Abbreviations: ISI, Insomnia Severity Index; M, mean; PSG, polysomnography; PSQI, Pittsburgh Sleep Quality Index; SD, standard deviation.

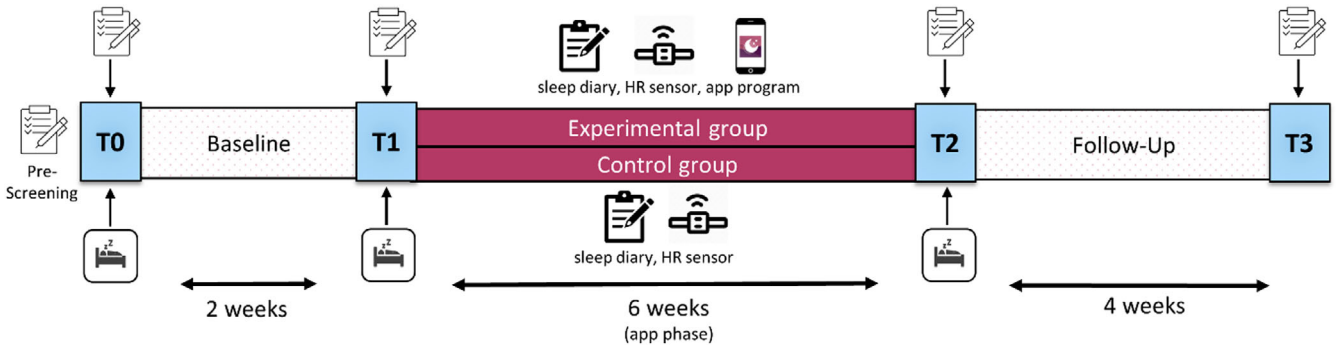


FIGURE 2 Study design. Effects of the CBT-I-based smartphone app (NUKKUAA®) on sleep were assessed subjectively via questionnaires and objectively via ambulatory polysomnography at the start of the study (T0), as well as the beginning (T1) and the end (T2) of the 6-week app phase. A follow-up assessment 4 weeks after the end of the app phase (T3) was conducted using questionnaires. Participants were randomly assigned to an experimental group (EG) or a control group (CG); both groups used a daily sleep diary, as well as a heart rate (HR) sensor (Polar®H10) for continuous objective sleep monitoring (T1–T2). During the app phase the EG used the app programme, receiving feedback and recommendations based on the recorded data, while the control group (CG) used the app for sleep monitoring only, without being provided with any feedback.

Electrodes were set up using the international 10–20 System (Klem et al., 1999) and included reference (Cz) and ground (Fz) electrodes as well as two frontal (F3, F4), two central (C3, C4), and two occipital (O1, O2) electrodes, as well as two mastoid electrodes for contralateral referencing (A1, A2). Eye movement was tracked using two EOG electrodes (placed 1 cm below the left outer canthus and 1 cm above the right outer canthus, respectively). Muscle activity was assessed using bipolar electromyography (EMG) on the chin. The electrocardiogram (ECG) signal was measured using two electrodes, placed near the right collarbone and on the lowest left rib.

PSG data were pre-processed as following: After contralateral re-referencing to mastoid electrodes a down sampling to 128 Hz as well

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as data-filtering using AASM recommendations (Bandpass-filter of 0.3–35 Hz for EEG and EOG, 10–100 Hz for EMG and 0.3–70 Hz for ECG; Iber et al., 2007) was performed using BrainVision Analyser 2.2 (Brain Products GmbH, Gilching, Germany). Afterwards, all files were manually checked for data quality, and if evaluated as "stageable" automatically sleep staged using the Sleepware G3 Software (Philips Respironics, Best, the Netherlands).

2.5 | Statistical analyses

Statistical analyses were performed using IBM SPSS Statistics, Version 29 (IBM Corp., 2021) and R.Studio, Version 2023.03.0 (R Core Team, 2018). Analyses were mainly based on parametric mixed-design ANOVAs, with the between-subject factor GROUP (EG vs. CG) and the within-subject factor TIME (T0, T1, T2 and T3 for subjective sleep analyses; T0, T1 and T2 for PSG analyses; weeks 1 and 6 of the app phase for discrepancy analyses).

In analyses regarding SOSD we assessed training-specific changes in the discrepancy of subjective (sleep diary-derived) and objective (HR sensor-derived) sleep data over the course of the app phase (T1–T2). Therefore, weekly averages of the according sleep parameter (subjective and objective TST, SE, SOL, and WASO, respectively) were calculated for each of the 6 weeks of the app phase. Missing values were replaced by the mean of all participants in the respective week. Next, the SOSDs were calculated for each week: TST, SE: subjective minus objective; SOL, WASO: objective minus subjective. A more positive discrepancy therefore describes a higher insomnia-related complaint, that is, an underestimation of TST and SE, and an overestimation of SOL and WASO.

Extreme outliers, that is, data points deviating more than ±2 SD from the mean, were excluded from the respective analysis. The statistical significance level was set to $p < 0.05$, while $p < 0.10$ are reported as statistical trends; effect sizes are taken into consideration for interpretation of practical significance. For measures of effect size, we provide partial eta square (η^2_{part}) and Cohen's *d*. In case of violations of sphericity, we applied a Greenhouse–Geisser or Huynh-Feldt correction. Bonferroni-corrected pairwise comparisons are reported. We hereby focussed on the relevant comparisons that include the app phase (i.e., between the beginning and the end of the app phase (T1–T2) and the beginning of the app phase and the follow-up assessment after 4 weeks (T1–T3) for ISI and PSQI, or T1–T2, and the beginning of the study and the end of the app phase (T0–T2), for PSG, respectively, as PSG was not included in the follow-up assessment). For the remaining pairwise comparisons and descriptive values of the outcome variables, refer to Tables S2–S6.

3 | RESULTS

3.1 | Subjective parameters

3.1.1 | Insomnia severity

Analysis revealed a significant effect of TIME (F (2.08, 107.93) = 38.34, $p < 0.001$, $\eta^2_{part} = 0.42$) and a significant interaction effect of TIME*GROUP (F (2.08, 107.93) = 6.14, $p = 0.003$, $\eta^2_{part} = 0.11$), yet no significant effect of GROUP (F (1, 52) = 1.86, $p = 0.179$, $\eta^2_{part} = 0.03$) for changes in insomnia severity (ISI; see Table 2 for results and Figure 3 a for visualisation).

The EG ($n = 27$) exhibited, as expected, no significant changes from T0–T1. However, the EG improved from T1–T2, and those improvements remained stable from T2–T3. Further significant improvements were found for T0–T2, T0–T3, and T1–T3. The CG ($n = 27$) showed a statistical trend for an improvement from T0–T1,

TABLE 2 Descriptive statistics and relevant pairwise comparisons for changes in severity of ISI and PSQI.

	T0	T1	T2	T3		
	M ± SD	M ± SD	M ± SD	M ± SD	T1–T2	T1–T3
ISI						
EG	15.48 ± 4.68	15.22 ± 4.86	10.22 ± 5.63	9.22 ± 5.57	$M_{Diff} = 5.00$, $CI_{95\%}$ [2.85, 7.15], $p < 0.001$, $d = 1.23$*	$M_{Diff} = 6.00$, $CI_{95\%}$ [3.48, 8.53], $p < 0.001$, $d = 1.25$*
CG	16.07 ± 5.19	14.70 ± 5.22	13.41 ± 4.92	12.52 ± 4.20	$M_{Diff} = 1.30$, $CI_{95\%}$ [−0.85, 3.45], $p = 0.624$, $d = 0.32$	$M_{Diff} = 2.19$, $CI_{95\%}$ [−0.34, 4.71], $p = 0.128$, $d = 0.46$
PSQI						
EG	9.07 ± 3.32	9.37 ± 3.70	6.56 ± 2.56	6.81 ± 3.66	$M_{Diff} = 2.82$, $CI_{95\%}$ [1.41, 4.22], $p < 0.001$, $d = 1.06$*	$M_{Diff} = 2.56$, $CI_{95\%}$ [1.00, 4.11], $p < 0.001$, $d = 0.87$*
CG	9.81 ± 3.34	9.67 ± 3.09	8.70 ± 3.33	8.56 ± 3.31	$M_{Diff} = 0.96$, $CI_{95\%}$ [-0.44, 2.37], $p = 0.391$, $d = 0.36$	$M_{Diff} = 1.11$, $CI_{95\%}$ [−0.44, 2.66], $p = 0.329$, $d = 0.38$

Note: Descriptive statistics and pairwise comparisons for insomnia severity (ISI) and subjective sleep quality (PSQI). Bonferroni corrected pairwise comparisons are reported. Here, relevant comparisons including the app phase are reported, see Table S2 for remaining pairwise comparisons. Statistically significant pairwise comparisons are marked with a star (*).

Abbreviations: CG, control group; EG, experimental group; ISI, Insomnia Severity Index; M, mean; PSQI, Pittsburgh Sleep Quality Index; SD, standard deviation; T0, start of the study; T1, beginning of app phase; T2, end of app phase; T3, follow-up after 4 weeks.

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no significant changes from T1–T2 and from T2–T3. However, significant improvements were found from T0–T2 and from T0–T3.

3.1.2 | Subjective sleep quality

Analysis revealed a significant effect of TIME (F (2.21, 114.79) = 17.94, $p < 0.001$, $\eta^2_{part} = 0.26$) and a statistical trend in the interaction effect of TIME*GROUP (F (2.21, 114.79) = 2.98, $p = 0.050$, $\eta^2_{part} = 0.05$), yet no significant effect of GROUP (F (1, 52) = 2.43, $p = 0.125$, $\eta^2_{part} = 0.05$), for changes in subjective sleep quality (PSQI; see Table 2 for results and Figure 3 b for visualisation). The EG ($n = 27$) exhibited, as expected, no significant changes from T0–T1. However, the EG improved from T1–T2, and those improvements remained stable from T2–T3. Further significant improvements were found for T0–T2, T0–T3, and T1–T3.

The CG ($n = 27$) showed no significant changes from T0–T1, from T1–T2, and from T2–T3.

3.2 | Objective sleep parameters

3.2.1 | Total sleep time

Analysis revealed a significant effect of TIME (F (2, 86) = 3.32, $p = 0.041$, $\eta^2_{part} = 0.07$) and a statistical trend towards an interaction effect of TIME*GROUP (F (2, 86) = 2.96, $p = 0.057$, $\eta^2_{part} = 0.06$), yet no significant effect of GROUP (F (1, 43) = 0.62, $p = 0.437$, $\eta^2_{part} = .0.01$), for changes in PSG-derived TST (see Table 3 for results and Figure 4 a for visualisation).

The EG ($n = 21$) exhibited no significant changes from T0–T1 and from T1–T2. The CG ($n = 24$) showed no significant changes

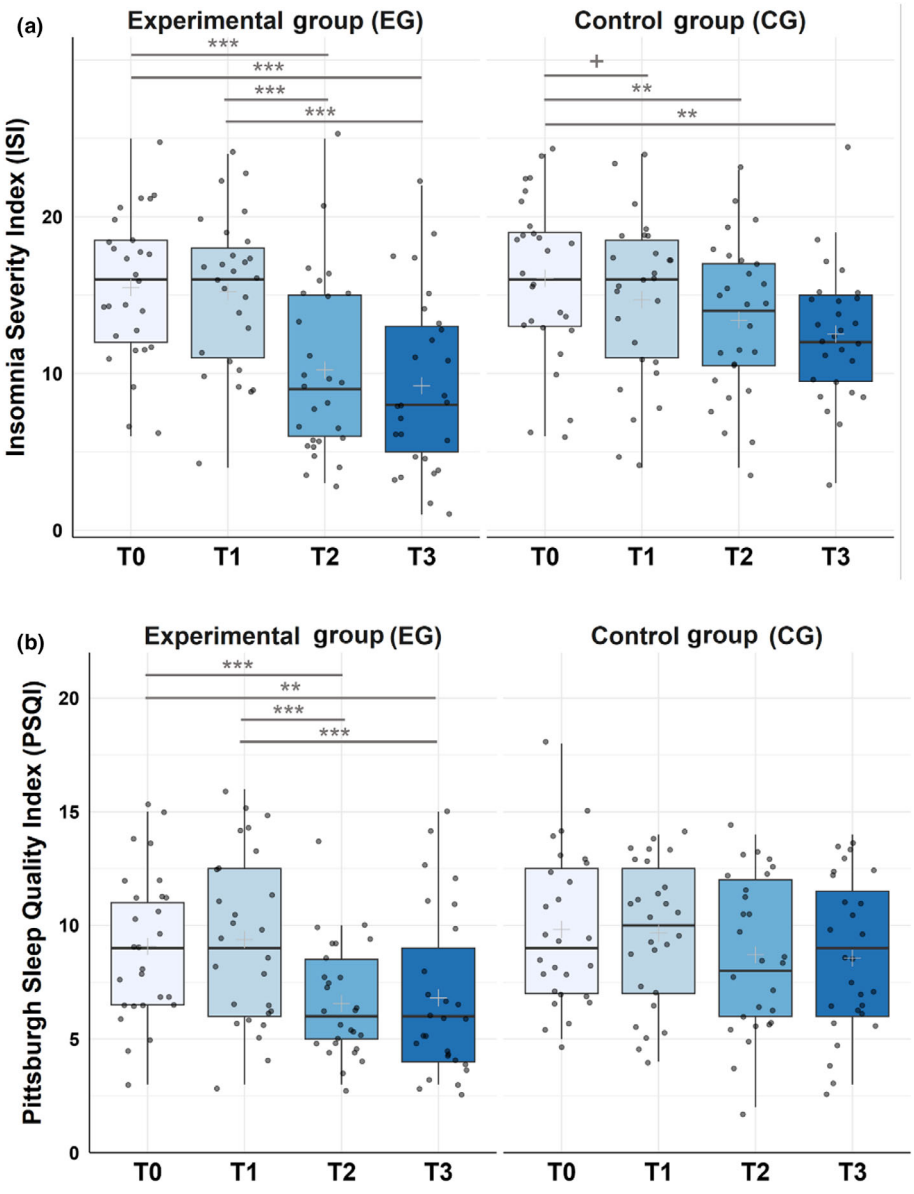


FIGURE 3 Changes in severity of insomnia symptoms and subjective sleep quality. (a) Insomnia severity, assessed with the Insomnia Severity Index (ISI) significantly improved from T0–T2, T0–T3, T1–T2, and T1–T3 (p's < 0.001) in the EG, and from T0–T2 ($p = 0.007$), and T0–T3 ($p = 0.002$) in the control group (CG). The CG further showed a trend towards an improvement from T0–T1 ($p = 0.091$). (b) Sleep quality, assessed with the Pittsburgh Sleep Quality Index (PSQI) significantly improved from T0–T2 ($p < 0.001$), T0–T3 ($p = 0.003$), T1–T2 ($p < 0.001$), and T1–T3 ($p < 0.001$) in the EG, The CG showed no significant changes ($p > 0.391$). Higher values represent a stronger impairment in the respective measure. Horizontal lines represent the medians, boxes the interquartile range, with whiskers depicting the 1.5 interquartile range. The grey cross corresponds to the mean. CG, control group; EG, experimental group; T0, start of the study; T1, beginning of app phase; T2, end of app phase; T3, follow-up after 4 weeks. Asterisks indicate significance: ***$p < 0.001$; **$p < 0.010$; +$p < 0.10$.

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from T0–T1 and from T1–T2, yet a significant reduction in TST from T0–T2.

3.2.2 | Time in bed

Analysis revealed a significant effect of TIME (F (2, 82) = 6.74, p = 0.002, η^2_{part} = 0.14) and a significant effect of GROUP (F (1, 41) = 4.57, p = 0.038, η^2_{part} = 0.10), yet no significant interaction effect of TIME*GROUP (F (2, 82) = 2.15, p = 0.123, η^2_{part} = 0.05), for changes in PSG-derived TIB (see Table 3 for results and Figure 4 b for visualisation).

The EG (n = 19) exhibited no significant changes from T0–T1 and from T1–T2. The CG (n = 24) showed no significant changes from T0–T1 and from T1–T2, yet a significant reduction in TIB from T0–T2.

3.2.3 | Sleep efficiency

Analysis revealed no significant effect of TIME (F (2, 86) = 0.51, p = 0.603, η^2_{part} = 0.01), no significant effect of GROUP (F (1, 43) = 1.19, p = 0.282, η^2_{part} = 0.03), and no significant interaction effect of TIME*GROUP (F (2, 86) = 0.91, p = 0.407, η^2_{part} = 0.02), for changes in PSG-derived SE. No pairwise comparisons between EG (n = 21) and CG (n = 24) are reported due to non-significance of main and interaction effects.

3.2.4 | Sleep onset latency

Analysis revealed no significant effect of TIME (F (2, 72) = 0.15, p = 0.864, η^2_{part} = 0.00), no significant effect of GROUP (F (1, 36) = 0.82, p = 0.370, η^2_{part} = 0.02), and no significant interaction effect of TIME*GROUP (F (2, 72) = 0.33, p = 0.723, η^2_{part} = 0.01) for changes in PSG-derived SOL. No pairwise comparisons between EG (n = 19) and CG (n = 19) are reported due to non-significance of main and interaction effects.

3.2.5 | Wake after sleep onset

Analysis revealed a significant effect of TIME (F (2, 80) = 5.89, p = 0.004, η^2_{part} = 0.13), yet no significant effect of GROUP (F (1, 40) = 1.55,

TABLE 3 Descriptive statistics and relevant pairwise comparisons for changes in PSG-derived sleep parameters.

	T0	T1	T2		
	M ± SD	M ± SD	M ± SD	T1–T2	T0–T2
TST					
EG	406.30 ± 62.15	417.31 ± 65.98	404.32 ± 39.69	M_{Diff} = 12.98, $CI_{95\%}$ [−19.19, 45.16], p = 0.961, d = 0.22	M_{Diff} = 1.98, $CI_{95\%}$ [−34.37, 38.32], p > 0.999, d = 0.03
CG	421.33 ± 62.58	396.53 ± 57.94	374.70 ± 79.33	M_{Diff} = 21.82, $CI_{95\%}$ [−8.28, 51.92], p = 0.234, d = 0.37	M_{Diff} = 46.63, $CI_{95\%}$ [12.63, 80.62], p = 0.004, d = 0.70*
TIB					
EG	496.18 ± 25.64	510.41 ± 74.25	480.11 ± 41.71	M_{Diff} = 30.30, $CI_{95\%}$ [−5.57, 66.17], p = 0.123, d = 0.48	M_{Diff} = 16.08, $CI_{95\%}$ [−18.77, 50.92], p = 0.768, d = 0.26
CG	485.62 ± 65.81	463.57 ± 59.61	436.83 ± 83.23	M_{Diff} = 26.74, $CI_{95\%}$ [−5.18, 58.66], p = 0.128, d = 0.43	M_{Diff} = 48.79, $CI_{95\%}$ [17.79, 79.79], p < 0.001, d = 0.80*
SE					
EG	82.47 ± 11.51	83.55 ± 10.52	85.45 ± 8.46	-	-
CG	86.78 ± 5.54	85.65 ± 6.71	86.14 ± 10.10	-	-
SOL					
EG	15.53 ± 10.29	17.58 ± 13.67	16.26 ± 11.68	-	-
CG	13.92 ± 9.36	13.58 ± 8.21	14.71 ± 6.14	-	-
WASO					
EG	63.68 ± 48.82	55.82 ± 40.50	42.66 ± 25.61	M_{Diff} = 13.16, $CI_{95\%}$ [−2.81, 29.13], p = 0.138, d = 0.47	M_{Diff} = 21.03, $CI_{95\%}$ [−0.73, 42.79], p = 0.061, d = 0.55
CG	49.72 ± 26.84	45.94 ± 30.40	35.24 ± 24.72	M_{Diff} = 10.70, $CI_{95\%}$ [−3.82, 25.21], p = 0.219, d = 0.38	M_{Diff} = 14.48, $CI_{95\%}$ [−5.30, 34.26], p = 0.224, d = 0.38

Note: Descriptive statistics and pairwise comparisons for PSG-derived sleep parameters TST, TIB, SE, SOL, and WASO. Bonferroni corrected pairwise comparisons are reported. Here, relevant comparisons including the app phase are reported, see Table S4 for remaining pairwise comparisons. For SE and SOL, pairwise comparisons are not reported due to non-significance of main and interaction effects. Statistically significant pairwise comparisons are marked with a star (*).

Abbreviations: CG, control group; EG, experimental group; M, mean; SD, standard deviation; SE, sleep efficiency; SOL, sleep onset latency; TIB, time in bed; TST, total sleep time; T0, start of the study; T1, beginning of app phase; T2, end of app phase; WASO, wake after sleep onset.

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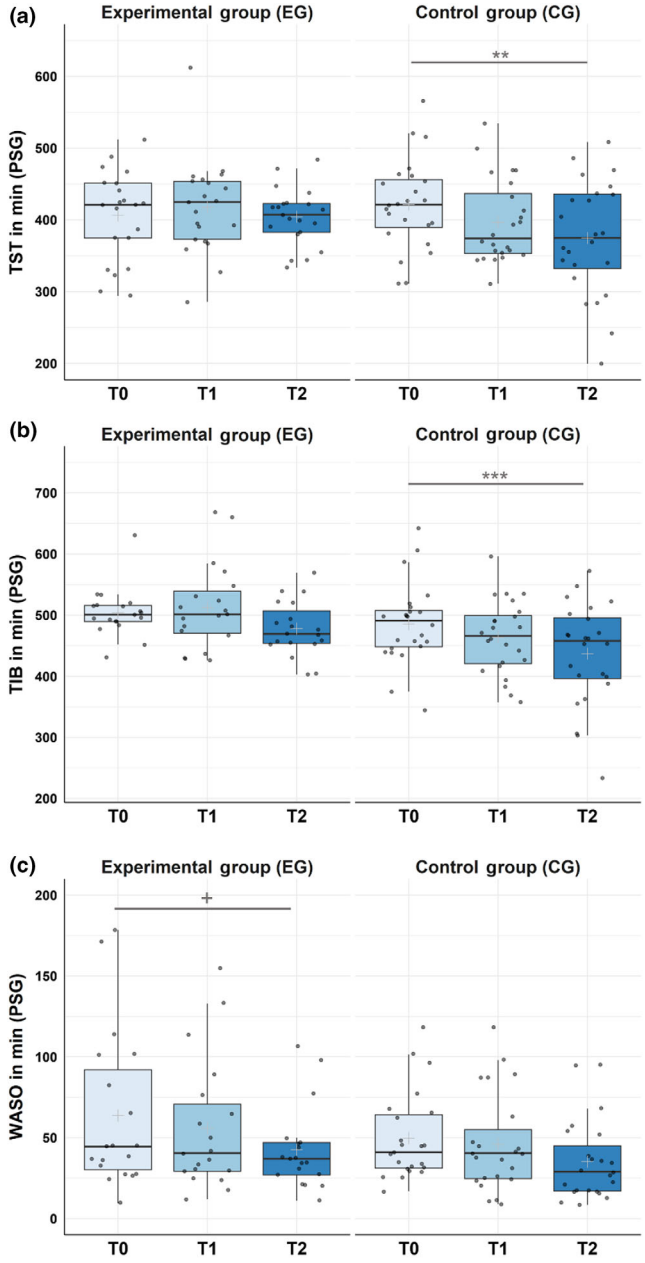


FIGURE 4 Changes in objective, PSG-derived sleep parameters. (a) TST did not significantly change for the EG ($p > 0.961$). However, TST significantly decreased from T0–T2 ($p = 0.004$) in the CG. (b) TIB did not significantly change for the EG ($p > 0.123$). However, TIB significantly decreased from T0–T2 ($p < 0.001$) in the CG. (c) WASO tendentially decreased from T0–T2 ($p = 0.061$) in the EG. The CG showed no significant changes ($p > 0.219$). Horizontal lines represent the medians, boxes the interquartile range, with whiskers depicting the 1.5 interquartile range. The grey cross corresponds to the mean. CG, control group; EG, experimental group; TIB, time in bed; TST, total sleep time; T0, start of the study; T1, beginning of app phase; T2, end of app phase; WASO, wake after sleep onset. Asterisks indicate significance: ***$p < 0.001$; **$p < 0.010$; +$p < 0.10$.

$p = 0.221$, $\eta^2_{part} = 0.04$), and no significant interaction effect of TIME*GROUP (F (2, 80) = 0.20, $p = 0.822$, $\eta^2_{part} = 0.01$) for changes in PSG-derived WASO (see Table 3 for results and Figure 4 c for visualisation).

The EG ($n = 19$) exhibited no significant changes from T0–T1 and from T1–T2, yet a statistical trend towards a reduction in WASO from T0–T2. The CG ($n = 23$) showed no significant changes from T0–T1 and from T1–T2.

3.3 | Subjective-objective sleep discrepancy

In the following analyses we assessed training-specific changes in the discrepancy of subjective (sleep diary-derived) and objective (sensor-derived) sleep data over the course of the app phase (T1–T2). A more positive discrepancy describes a higher insomnia-related complaint, that is, an underestimation of TST and SE, and an overestimation of SOL and WASO (see Table 4 for full results and Figure 5 for visualisation).

3.3.1 | Total sleep time

Analysis revealed a significant effect of TIME (F (1, 52) = 10.84, $p = 0.002$, $\eta^2_{part} = 0.17$), yet no significant effect of GROUP (F (1, 52) = 0.08, $p = 0.775$, $\eta^2_{part} = 0.00$), and no significant interaction effect of TIME*GROUP (F (1, 52) = 0.03, $p = 0.855$, $\eta^2_{part} = 0.00$), for changes in SOSD of TST. Both groups, the EG ($n = 27$) and the CG ($n = 27$), showed significant improvements towards less underestimation of TST from week 1 to 6 (cf. Figure 5 a).

3.3.2 | Sleep efficiency

Analysis revealed a significant effect of TIME (F (1, 52) = 4.85, $p = 0.032$, $\eta^2_{part} = 0.09$), yet no significant effect of GROUP (F (1, 52) = 0.00, $p = 0.959$, $\eta^2_{part} = 0.00$), and no significant interaction effect of TIME*GROUP (F (1, 52) = 1.45, $p = 0.233$, $\eta^2_{part} = 0.03$), for changes in SOSD of SE. In contrast to the CG ($n = 28$), the EG ($n = 26$) exhibited significant improvements towards less underestimation of SE from week 1 to 6 (cf. Figure 5 b).

3.3.3 | Sleep onset latency

Analysis revealed a significant effect of TIME (F (1, 52) = 4.90, $p = 0.031$, $\eta^2_{part} = 0.09$), yet no significant effect of GROUP (F (1, 52) = 0.95, $p = 0.335$, $\eta^2_{part} = 0.02$), and no significant interaction effect of TIME*GROUP (F (1, 52) = 1.54, $p = 0.220$, $\eta^2_{part} = 0.03$), for changes in SOSD of SOL. In contrast to the CG ($n = 28$), the EG ($n = 26$) exhibited significant improvements towards less overestimation of SOL from week 1 to 6. (cf. Figure 5 c).

3.3.4 | Wake after sleep onset

Analysis revealed a significant effect of TIME (F (1, 51) = 9.37, $p = 0.004$, $\eta^2_{part} = 0.16$), yet no significant effect of GROUP (F (1,

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TABLE 4 Descriptive statistics and pairwise comparisons for changes in in discrepancy between subjective (sleep diary-derived) and objective (HR sensor-derived) sleep parameters during the app phase.

	Week 1	Week 6	
	M ± SD	M ± SD	Weeks 1–6
TST			
EG	13.63 ± 56.93	−6.80 ± 27.07	$M_{Diff} = 20.43$, $CI_{95\%}$ [3.75, 37.10], $p = 0.017$, $d = 0.47$*
CG	10.04 ± 32.81	−8.22 ± 30.16	$M_{Diff} = 18.26$, $CI_{95\%}$ [1.58, 34.94], $p = 0.032$, $d = 0.42$*
SE			
EG	4.08 ± 12.86	−1.49 ± 5.46	$M_{Diff} = 5.57$, $CI_{95\%}$ [0.85, 10.29], $p = 0.022$, $d = 0.46$*
CG	2.21 ± 8.28	0.58 ± 9.25	$M_{Diff} = 1.63$, $CI_{95\%}$ [−2.92, 6.18], $p = 0.476$, $d = 0.14$
SOL			
EG	4.18 ± 17.30	−3.88 ± 7.19	$M_{Diff} = 8.05$, $CI_{95\%}$ [1.32, 14.79], $p = 0.020$, $d = 0.47$*
CG	4.97 ± 20.90	2.71 ± 16.32	$M_{Diff} = 2.26$, $CI_{95\%}$ [−4.23, 8.75], $p = 0.487$, $d = 0.13$
WASO			
EG	−7.78 ± 40.97	−29.75 ± 28.27	$M_{Diff} = 21.98$, $CI_{95\%}$ [8.07, 35.89], $p = 0.003$, $d = 0.61$*
CG	−12.08 ± 19.11	−20.38 ± 17.23	$M_{Diff} = 8.31$, $CI_{95\%}$ [−5.87, 22.48], $p = 0.245$, $d = 0.22$

Note: Descriptive statistics and pairwise comparisons for changes in discrepancy between subjective (sleep diary-derived) and objective (HR sensor-derived) sleep parameters during the 6-week app phase for TST, SE, SOL, and WASO. Bonferroni corrected pairwise comparisons are reported. Statistically significant pairwise comparisons are marked with a star (*).
Abbreviations: CG, control group; EG, experimental group; M, mean; SD, standard deviation; SE, sleep efficiency; SOL, sleep onset latency; TST, total sleep time; week 1, first week of the 6-week app phase; week 6, sixth/last week of the 6-week app phase; WASO, wake after sleep onset.

51) = 0.18, $p = 0.672$, $\eta^2_{part} = 0.00$), and no significant interaction effect of TIME*GROUP (F (1, 51) = 1.91, $p = 0.173$, $\eta^2_{part} = 0.04$), for changes in SOSD of WASO. In contrast to the CG ($n = 28$), the EG ($n = 27$) exhibited a significant reduction of the existing underestimation of WASO from week 1 to 6 (cf. Figure 5 d).

4 | DISCUSSION

The present study investigated the efficacy of an app-based dCBT-I programme on subjective and objective measures of sleep as well as changes in SOSD. While subjective effects of dCBT-I are well studied, to our knowledge no RCT has investigated the influence of an app-based dCBT-I programme on objective measures of sleep and on SOSD so far.

Regarding subjective outcomes of insomnia severity and subjective sleep quality, the results indicate training-specific improvements, as we found a significant decrease of mean scores in ISI and PSQI and large effect sizes during the app phase for the EG, yet not for the CG. These improvements of the EG remained stable during a 4-week follow-up period. The CG showed medium effect sizes regarding improvements in insomnia severity, however, those improvements mainly occurred over the course of the whole study, making them less specific to the app phase and therefore serving as an indicator for the efficacy of the NUKKUAA® app programme. The attained results are in line with research examining other dCBT-I programmes, yielding large effect sizes regarding improvements in subjective outcomes (e.g., Lorenz et al., 2019; Ritterband et al., 2017).

Besides statistical improvements, it is further necessary to investigate the clinical impact of the app programme. On average, the EG improved from 15.22 to 10.22 points on the ISI during training, indicating a change from "moderate clinical insomnia" (defined by an ISI-range of 15–22) to "sub-threshold insomnia" (defined by an ISI-range of 8–14).

Following the recommendation of Yang et al. (2009) and defining the minimally important difference (MID) in ISI with a change of 6 points, we found that 12 participants of the EG (44.4%) can be labelled as responders from the beginning (T1) to the end of the app phase (T2), and to the follow-up (T3), respectively. Nine of the responders at T2 remained a responder at T3, while three further participants fulfilled the MID at T3. This suggests that overall, 15 of 27 participants in the EG (55.5%) can be labelled as responders at one point in the study. In the CG four (14.8%) and five (18.5%) participants achieved a MID of 6 points from T1 to T2 or T3, respectively.

Regarding subjective sleep quality, the EG improved from 9.37 to 6.56 points on the PSQI during training, both values are inside the range of "poor sleep quality" (defined by a PSQI-range of 6–10). Thus, the EG on average did not achieve a categorical change.

Regarding objective, PSG-derived outcomes we found changes in WASO, TST, and TIB over the course of the study when considering all participants. For WASO, this effect seems to be driven by a tendential decrease in wake time in the EG from the beginning of the study to the end of the app phase, which might be due to a reduction of variance to overall lower levels of WASO from T0 to T2. For TST and TIB, however, this effect is more likely attributed to a reduction of these parameters in the CG. It is, however, to be noted, that

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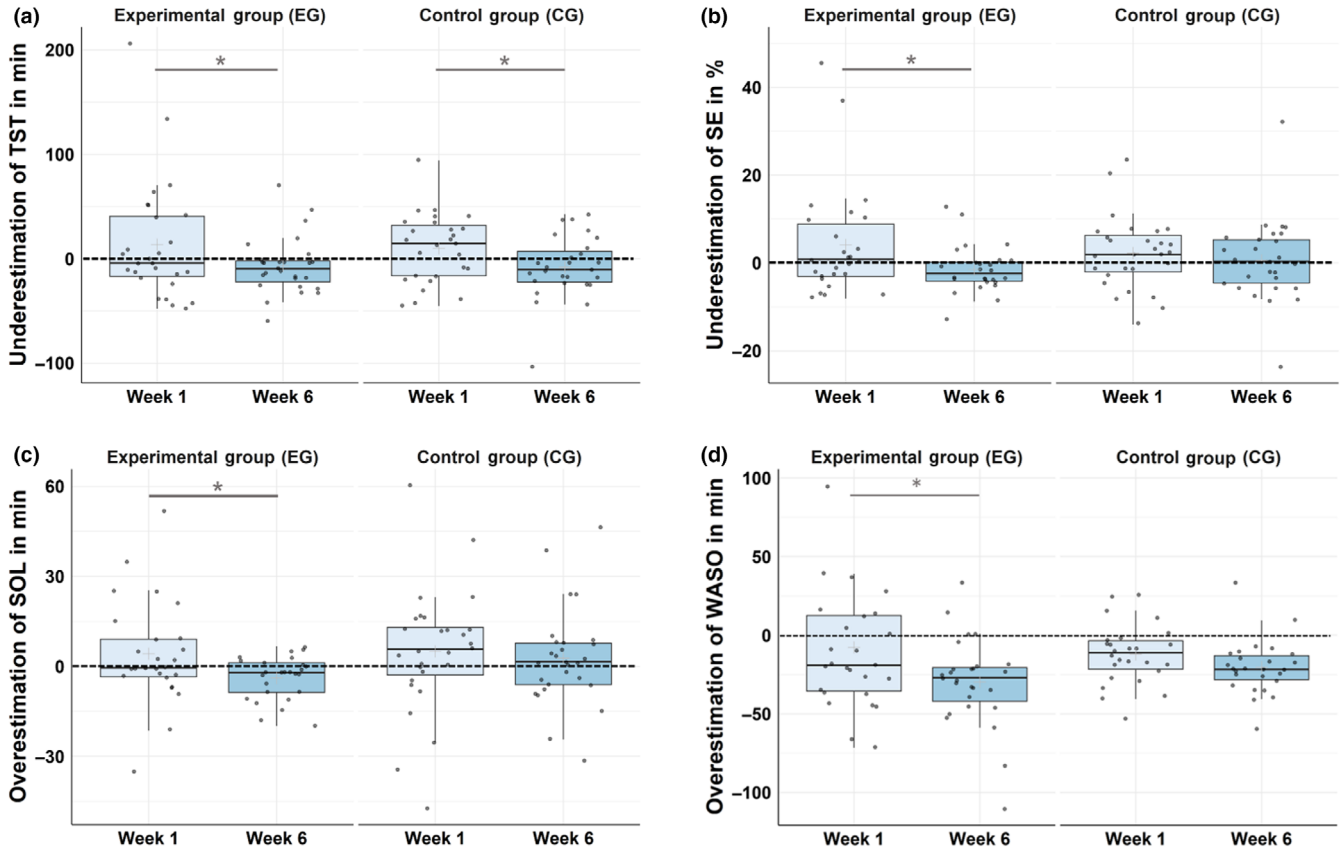


FIGURE 5 Changes in discrepancy between subjective (sleep diary-derived) and objective (HR sensor-derived) sleep parameters during the app phase. (a) Underestimation of TST in minutes significantly improved in both EG ($p = 0.017$) and CG ($p = 0.032$). (b) Underestimation of SE in percent significantly improved in the EG ($p = 0.022$), yet not in the CG ($p = 0.476$). (c) Overestimation of SOL in minutes significantly improved in the EG ($p = 0.020$), yet not in the CG ($p = 0.487$). (d) Already existing underestimation in WASO in minutes significantly decreased in the EG ($p = 0.003$), yet not in the CG ($p = 0.245$). Horizontal lines represent the medians, boxes the interquartile range, with whiskers depicting the 1.5 interquartile range. The grey cross corresponds to the mean. CG, control group; EG, experimental group; SE, sleep efficiency; SOL, sleep onset latency; TST, total sleep time; WASO, wake after sleep onset; week 1, first week/beginning of app phase; week 6, last week/end of app phase. Asterisks indicate significance: ***$p < 0.001$; **$p < 0.010$; *$p < 0.050$.

interaction effects were not statistically significant (for TIB and WASO) or displayed a statistical trend only (for TST), and therefore follow-up pairwise comparisons of these analyses should be interpreted with caution and serve more as an insight in potential changes over the course of the study. Objectively assessed SOL and SE on the other hand did not change significantly during the study.

When investigating these findings, it is important to address the initially measured values to understand whether significant changes were likely to happen in the first place, especially when including a sleep-disturbed, yet non-clinical sample, as in the present study. For PSG-derived SOL, at the beginning of the study the EG and the CG needed on average 15.53 and 13.92 min to fall asleep, respectively. Similarly, SE was at 82.47% in the EG and 86.78% in the CG. These numbers are inside or closer to the range of healthy sleep than common complaints of patients with primary insomnia (e.g., SOL: 20.0 ± 17.1, SE: 81.8 ± 10.5; Baglioni et al., 2014), which makes significant improvements relatively unlikely to happen. However, this is not a rare occurrence in PSG research. Mitchell et al. (2019) noticed that most studies examining PSG outcomes, based their recruitment of participants on subjective sleep data rather than on objective assessment using PSG, as also done in the present study. While this is comprehensible due to feasibility and given the subjective definition of insomnia lacking objective criteria, it might not reflect all relevant aspects of sleep problems. Future studies examining objective outcomes of sleep should consider a screening using objective methods in order to avoid ceiling effects of sleep. Given high costs for PSG, increasing the application of accurate sleep tracking by the use of validated algorithms could represent a suitable approach in this regard.

Nevertheless, the presence of subjective sleep complaints in people with objectively measured "normal" sleep is a common pattern in insomnia sufferers (i.e., SOSD). To our knowledge, this is the first study investigating the effects of dCBT-I on SOSD using HR sensor-derived data in comparison with sleep diary data. Participants showed the typical image of an overestimation of SOL, as well as an underestimation of TST and SE. Similar to our results regarding PSG-derived data, interaction effects for SOSD did not reach statistical significance, which, again, implies, that the following interpretations rather

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describe a tendency of study-related outcomes: In line with Kay et al. (2015) and Nishikawa et al. (2021), we found an amelioration in the overestimation of SOL and the initial underestimation of SE in the EG only, both showing close to medium effect sizes. Regarding the initial underestimation of TST, we found a decrease in both groups, even though the CG was not provided with any feedback regarding their sleep. However, participants in the CG still might have been able to improve their estimation of TST during the 6-week interval due to self-observation based on daily sleep diaries, and therefore generally more awareness of their sleep. This might be observed only in the SOSD of TST, as the other parameters (i.e., SOL, SE, WASO) might be even harder to estimate for the individual, wherefore a correction regarding those parameters seems not to take place without corresponding feedback.

Interestingly and contrary to previous findings concerning WASO, we observed an underestimation in both groups already at the beginning of the app phase. This could be explained by a possibly misleading question regarding WASO in the sleep diary, resulting in participants not including wake time in bed after their final awakening, while this was part of the calculation of sensor-derived WASO. Furthermore, an underestimation of WASO on average may not be surprising since short awakenings in the night are typically not recalled, however, interestingly, it also seems to be found in insomnia patients (Valko et al., 2021). Nevertheless, the underestimation significantly increased during the app phase only in the EG, which can be interpreted as a healthier, somewhat positively biased perception of their sleep. Moreover, as SOSD seems to be related to WASO (Knab & Engel, 1988), the tendential reduction of PSG-derived WASO in the EG might lead to an improved subjective sleep quality, which in turn might be associated with a more accurate estimation concerning subjective sleep continuity parameters, and consequently SOSD.

In general, we assume that providing feedback to the EG daily during the app phase increases the awareness of the differences between subjective estimation and objective measurement of sleep, which might explain the more accurate sleep perception by the end of the app phase. (Tang & Harvey, 2004). In particular, written feedback might be important in this regard (Tang & Harvey, 2006).

Generally, providing feedback on sleep seems to be beneficial regarding insomnia severity and sleep disturbances in individuals with significant insomnia symptoms (Spina et al., 2023). However, it is also important to note, that despite potential improvements in sleep parameters and SOSD, it is not yet fully understood, whether reporting feedback on objective sleep might also possibly result in negative effects, for example, an increase of obsession with sleep due to selected attention, a key pathological feature of insomnia. This is one important reason for having a valid and reliable objective sleep measurement, to prevent potential negative effects due to false or misleading feedback (Gavriloff et al., 2018). In the present app-based programme, such valid sleep measurement is implemented (see Topalidis, Baron, et al., 2023; Topalidis, Heib, et al., 2023). Since the literature on effects of (objective) sleep feedback on insomnia, especially in combination with dCBT-I, is still sparse, further research is needed in order to allow for reliable statements regarding feedback on sleep in insomnia patients.

4.1 | Limitations

As in the present study a sleep-disturbed, yet non-clinical, sample was included (i.e., ISI-score ≥8 or PSQI-score ≥6 at pre-screening, no requirement for a diagnosis of insomnia disorder), it should be noted, that especially for objectively derived sleep measures, significant improvements were less likely to be expected. Furthermore, during the 6-week app phase only five participants of the EG (17.8%) completed all 6 levels of the sleep training, which were required to experience core contents of CBT-I; a total of 14 participants (50%) completed level 5. Since for the completion of a level it was required to listen to the relaxation exercise 7 times, which in the present study design was equal to listening to it every day in order to complete the core app programme in the 6-week app phase, skipping a couple of days led to a delay in progression. For further studies, this requirement should either be reduced, or the length of the app phase prolonged. Furthermore, it should be mentioned that the application of the HR sensor made some, especially elderly, participants unsure regarding potential technical problems, for example, connection issues between the sensor and the smartphone, which might have also affected their sleep.

5 | CONCLUSION

The CBT-I-based app programme NUKKUAA® appears to be feasible for reducing insomnia severity and improving subjective sleep quality. It might also allow users to perceive aspects of their sleep more accurately and therefore enable them to become aware of their improvements during training more easily. While no training-specific effects were found on objective sleep so far, including larger samples containing more insomniacs suffering from objective sleep impairments, should be investigated further.

Our findings indicate the beneficial effects of a novel smartphone app on sleep as well as on SOSD. This study is part of a growing literature investigating app-based programmes, aiming at deepening the understanding of effects of digital programmes on both subjective and objective aspects of sleep. Access to such programmes should be guaranteed and may help to reduce the gap in care.

AUTHOR CONTRIBUTIONS

Alexandra Hinterberger: Conceptualization; writing – original draft; investigation; methodology; visualization; writing – review and editing; formal analysis; project administration. **Esther-Sevil Eigl:** Conceptualization; writing – original draft; investigation; methodology; visualization; writing – review and editing; formal analysis. **Robyn Nina Schwemlein:** Writing – original draft; investigation. **Pavlos Topalidis:** Visualization; formal analysis. **Manuel Schabus:**

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Conceptualization; writing – review and editing; methodology; supervision; project administration.

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CONFLICT OF INTEREST STATEMENT

Manuel Schabus is Co-Founder and CSO of NUKKUAA®. AH and E-SE received an honorarium for data collection from the NUKKUAA GmbH.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Alexandra Hinterberger https://orcid.org/0000-0001-5043-9909
Esther-Sevil Eigl https://orcid.org/0000-0002-0018-7632
Robyn Nina Schwemlein https://orcid.org/0009-0000-9533-8597
Pavlos Topalidis https://orcid.org/0000-0002-6235-060X
Manuel Schabus https://orcid.org/0000-0001-5899-8772

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